

RESEARCH PROFILE

Robotics Engineering Ph.D. student at Worcester Polytechnic Institute and member of the ELPIS Lab. Research interests include motion and kinodynamic planning, robot learning, nonprehensile manipulation, adaptive planning, and planning under uncertainty.

EDUCATION

Aug 2024–present	Ph.D. in Robotics Engineering Worcester Polytechnic Institute · ELPIS Lab GPA 4.0/4.0 across the first 36 credits.
Sep 2018–Sep 2023	B.S. in Mechanical Engineering Iran University of Science and Technology GPA 16.12/20 (3.34/4.0).
Sep 2014–Jul 2018	High School Diploma · Mathematics and Physics National Organization for Development of Exceptional Talents (NODET) GPA 19.01/20 (4.0/4.0).

RESEARCH EXPERIENCE

Aug 2024–present	Research Assistant · Learning Dynamics for Manipulation and Motion Planning ELPIS Lab, Worcester Polytechnic Institute Kinodynamic planning and nonprehensile manipulation; UR10/UR16e trajectory control and learned physics residuals.
Sep 2021–Jul 2024	Research Assistant · Simulation and Construction of Legged Robots Mechatronics Laboratory, Iran University of Science and Technology Legged-robot simulation, construction, and control.

SELECTED ENGINEERING PROJECTS AND INTERNSHIP

- Unitree G1 humanoid development and control (Jan 2026); OpenManipulator X robotics control (Nov 2024).
- B.S. dissertation: quadruped-robot simulator (Oct 2021–Sep 2023).
- Mechanical Engineering Intern, Rahe Andisheh Company (Jul–Sep 2020): reverse-engineered a shrimp-pond oxygenation machine; designed chassis, selected gearbox and actuator components, and supported structural analysis, manufacturing, and testing.

TEACHING EXPERIENCE

Aug 2025–May 2026	Teaching Assistant · Motion Planning and Machine Learning for Robotics Worcester Polytechnic Institute
Jan–May 2025	Teaching Assistant · Mechanical Applications in Robotics; Sensing and Perception in Robotics Worcester Polytechnic Institute
Sep 2023–Feb 2024	Teaching Assistant · Fundamentals of Computer Programming Iran University of Science and Technology
Sep 2022–Feb 2023	Teaching Assistant · Vectorial Dynamics and Dynamics of Machinery Iran University of Science and Technology
Sep 2021–May 2024	Physics Teacher NODET High School

SELECTED PUBLICATIONS

AURA: Asymptotically Optimal Uncertainty-Robust Replanning Algorithm for Kinodynamic Systems

Seyedali Golestaneh, Zhuoyun Zhong, Donghyung Lee, and Constantinos Chamzas
IEEE Robotics and Automation Letters, 2026 · Accepted September 2026 · Paper

MetaPusher: Meta Learning and Planning for Nonprehensile Manipulation of Unseen Objects with Rapid Online Adaption

Donghyung Lee, Seyedali Golestaneh, Jaskrit Singh, Zhuoyun Zhong, Athanasios Kapoutsis, and Constantinos Chamzas
arXiv preprint, 2026 · Co-first author · Paper

Terminal Matters: Kinodynamic Planning with a Terminal Cost and Learned Uncertainty in Belief State-Cost Space

Zhuoyun Zhong, Seyedali Golestaneh, and Constantinos Chamzas
arXiv preprint, 2026 · Paper

ActivePusher: Active Learning and Planning with Residual Physics for Nonprehensile Manipulation

Zhuoyun Zhong, Seyedali Golestaneh, and Constantinos Chamzas
IEEE International Conference on Robotics and Automation (ICRA), 2026 · Best Student Paper Award · Paper

Robust and Efficient Phase Estimation in Legged Robots via Signal Imaging and Deep Neural Networks

Kamyab Yazdipaz, Nooshin Kohli, Seyed Ali Golestaneh, and Mohammad Shahbazi
IEEE Access, vol. 13, pp. 49018–49029, 2025 · Paper

AWARDS AND RECOGNITION

2026	Best Student Paper Award IEEE International Conference on Robotics and Automation (ICRA)
2026	Best Student Paper Award Hellenic Robotics Forum (HRF)
2026	Glenn Yee Travel Award Robotics Engineering Graduate Student, WPI
2022	Top 10% among Mechanical Engineering students Iran University of Science and Technology
2018	Top 0.75% among 144,437 participants in the National University Entrance Exam Iran
2014	4th place · Soccer 2D Simulation RoboCup Iran Open International Competitions

TECHNICAL SKILLS AND LANGUAGES

Programming: C++, Python, MATLAB, EES. Robotics and simulation: ROS, Genesis, MuJoCo, Arduino. Engineering: SolidWorks, CATIA, ADAMS. Tools: Linux, Windows, LaTeX, Microsoft Office, Vim.
Languages: Persian (native), English (fluent).